Project: Worked examples
Title: Motor and actuator sizing
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Torque, speed and power a motor needs to move a load through a mechanism on a point-to-point cycle: the load reflected to the motor shaft, the acceleration from the motion profile, peak torque during acceleration, RMS torque over the cycle (thermal duty), peak speed, inertia ratio, and a check of a candidate motor's rated and peak torque and speed with a safety margin. A rotary load that runs continuously (a pump, a fan) is sized instead from its steady torque and speed and a run-up time. Linear mechanisms: k = metres per motor radian (screw p/(2π i); belt D/(2 i); rack D_p/(2 i)); J_refl = m k²/η; T_load = F k/η with F = m g (sin θ + μ cos θ) + F_p. Rotary: J_refl = J_L/(i² η), T_load = T_L/(i η). Trapezoidal 1/3–1/3–1/3 (adjustable): v_pk = s/(t_move − t_acc), a = v_pk/t_acc; S-curve: sine acceleration with a_pk = (π/2) a_trap and the same v_pk. T_acc = J_total α; T_pk = T_acc + T_load; T_rms = √(Σ T_i² t_i / t_cycle) including the dwell. RMS mechanical power (the thermal-equivalent continuous power a supply sees; regeneration counts as power): P_rms = ω_pk √{[t_acc·(2/3)(T_load² + T_acc²) + T_load² t_const]/t_cycle} (trapezoid), with t_acc·[(3/4) T_load² + (5/16) T_acc²] for the S-curve; the instantaneous peak P_pk is not a continuous demand. Continuous rotary duty: ω = 2π n_L i/60; T_load = T_L/(i η); T_rms = T_load; T_pk = J_total ω/t_start + T_load (full load during the run-up, conservative); n_pk = n_L i; P_cont = T_load ω; P_rms = P_cont; P_rated = T_rated·2π n_rated/60, the candidate motor's nameplate rated shaft power from its rated torque and rated speed n_rated (not the running speed), the shaft power its supply must be sized for; without n_rated there is no P_rated and the sheet warns.

Ref: Hughes & Drury, Electric Motors and Drives Ch. on motor and drive selection — Load torque, inertia referred to the motor, duty cycles
Ref: Manufacturer servo-sizing guides (Kollmorgen, Parker) — Reflected inertia, RMS torque, inertia ratio practice

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mechanism	Mechanism	`lead / ball screw`		assumption: Sizing convention.	any
i	Gear ratio i (motor : output, 1 = none)	`1`		assumption: Sizing convention.	1 … 1000
eta	Mechanism efficiency (overall)	`0.9`		spec: Screw datasheet BS-1610: 90 %	0.3 … 1
p_screw	Screw lead (per revolution of the screw)	`10`	mm	spec: Screw datasheet BS-1610: 10 mm lead	1 … 100 mm
J_mech	Transmission inertia referred to the motor shaft (screw, pulleys, gearbox, coupling)	`0.0001`	kg*m^2	spec: Screw datasheet BS-1610 + coupling: 1.0e-4 kg·m²	0 … 10 kg*m^2
m	Moved mass (linear mechanisms)	`50`	kg	spec: Mechanical layout ML-12: carriage + gripper 50 kg	0 … 100000 kg

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
mu	Friction coefficient (guide / bearing)	0.1		assumption: Linear guides with seals: μ = 0.1.	0 … 1
theta	Incline from horizontal (+ moving uphill)	0	deg	assumption: Sizing convention.	-90 … 90 deg
F_p	Process force opposing the move (linear)	0	N	assumption: Sizing convention.	0 … 1000000 N
T_L	Constant load torque at the output (rotary)	0	N*m	default	0 … 100000 N*m
duty_mode	Duty (point-to-point move cycle, or continuous rotary running with a run-up)	point-to-point		default	any
profile	Motion profile	trapezoidal		assumption: Sizing convention.	any
s	Move distance (linear: m; rotary: output revolutions)	0.5		spec: Cycle specification CS-3: 0.5 m in 0.5 s, 0.5 s dwell, 1/3 ramps	0.0001 … 10000
t_move	Move time	0.5	s	spec: Cycle specification CS-3: 0.5 m in 0.5 s, 0.5 s dwell, 1/3 ramps	0.01 … 3600 s
f_acc	Acceleration (and deceleration) fraction of the move time	0.333333		spec: Cycle specification CS-3: 0.5 m in 0.5 s, 0.5 s dwell, 1/3 ramps	0.05 … 0.5
t_dwell	Dwell between moves	0.5	s	spec: Cycle specification CS-3: 0.5 m in 0.5 s, 0.5 s dwell, 1/3 ramps	0 … 3600 s
J_motor	Candidate motor rotor inertia	0.0002	kg*m^2	spec: Motor datasheet SM-040: 2.0e-4 kg·m², 2.0 / 6.0 N·m, 10 000 rpm	1e-7 … 10 kg*m^2
T_rated	Candidate motor continuous (rated) torque	2	N*m	spec: Motor datasheet SM-040: 2.0e-4 kg·m², 2.0 / 6.0 N·m, 10 000 rpm	0.01 … 100000 N*m
T_peak	Candidate motor peak torque	6	N*m	spec: Motor datasheet SM-040: 2.0e-4 kg·m², 2.0 / 6.0 N·m, 10 000 rpm	0.01 … 100000 N*m

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
n_max	Candidate motor maximum speed	10000	rpm	spec: Motor datasheet SM-040: 2.0e-4 kg·m², 2.0 / 6.0 N·m, 10 000 rpm	10 … 100000 rpm
SF	Safety margin on torque	1.25		assumption: Sizing convention.	1 … 3
J_ratio_max	Inertia ratio limit (load : motor), warning	10		assumption: Sizing convention.	1 … 100

3. Assumptions

A1. Rigid transmission; the mechanism efficiency applies to the driving direction and is also used in the deceleration phase (conservative for peak, slightly pessimistic for RMS). (Servo-sizing practice.)
A2. The friction force is Coulomb friction on the moved mass; viscous and breakaway friction are not modelled unless included in F_p. (Load model.)
A3. Continuous duty: the load torque is constant at the stated speed and the motor accelerates the whole inertia linearly to speed in t_start while carrying the full load torque (conservative); the move distance, move time, acceleration fraction, dwell and profile are ignored. (Run-up idealisation for pumps and fans.)
A4. The S-curve profile has sine-shaped acceleration over the same acceleration time, so its peak acceleration is π/2 times the trapezoidal value. (Jerk-limited profile idealisation.)
A5. mechanism (Mechanism) is taken as lead / ball screw. (Sizing convention.)
A6. i (Gear ratio i (motor : output, 1 = none)) is taken as 1. (Sizing convention.)
A7. mu (Friction coefficient (guide / bearing)) is taken as 0.1. (Linear guides with seals: μ = 0.1.)
A8. theta (Incline from horizontal (+ moving uphill)) is taken as 0 deg. (Sizing convention.)
A9. F_p (Process force opposing the move (linear)) is taken as 0 N. (Sizing convention.)
A10. profile (Motion profile) is taken as trapezoidal. (Sizing convention.)
A11. SF (Safety margin on torque) is taken as 1.25. (Sizing convention.)
A12. J_ratio_max (Inertia ratio limit (load : motor), warning) is taken as 10. (Sizing convention.)

4. Working

```
[22] k = p_screw / (2 * pi * i)
= 10 mm / (2 × π × 1)
= 1.5915 mm
```
Travel per motor radian
```
[23] F_load = m * 9.80665 m/s^2 * (sin(theta) + mu * cos(theta)) + F_p
= 50 kg × 9.80665 m/s^2 × (sin(0 deg) + 0.1 × cos(0 deg)) + 0 N
= 49.033 N
```
Load force along the motion (gravity, friction, process)
```
[24] T_load = F_load * k / eta
= 49.033 N × 1.5915 mm / 0.9
= 0.08671 N·m
```
Load torque at the motor shaft
```
[25] J_refl = m * k^2 / eta
= 50 kg × (1.5915 mm)^2 / 0.9
= 0.00014072 kg·m²
```
Load inertia reflected to the motor shaft
```
[26] v_pk = s * 1 m / (t_move * (1 - f_acc))
= 0.5 × 1 m / (0.5 s × (1 - 0.33333))
= 1.5 m/s
```
Peak linear velocity (trapezoid)

[27] a_trap = v_pk / (f_acc * t_move)
= 1.5 m/s / (0.33333 × 0.5 s)
= 9 m/s²
Trapezoidal acceleration
[28] omega_pk = v_pk / k
= 1.5 m/s / 1.5915 mm
= 942.48 rad/s
Peak motor speed
[30] a_pk = a_trap
= 9 m/s²
= 9 m/s²
Peak acceleration
[31] alpha_pk = a_pk / k
= 9 m/s² / 1.5915 mm
= 5654.9 rad/s²
Peak motor angular acceleration
[32] J_total = J_motor + J_mech + J_refl
= 0.0002 kg·m² + 0.0001 kg·m² + 0.00014072 kg·m²
= 0.00044072 kg·m²
Total inertia at the motor shaft
[33] T_acc = J_total * alpha_pk
= 0.00044072 kg·m² × 5654.9 rad/s²
= 2.4922 N·m
Acceleration torque
[34] T_pk = T_acc + T_load
= 2.4922 N·m + 0.08671 N·m
= 2.5789 N·m
Peak torque (accelerating against the load)
[35] T_dec = -T_acc + T_load
= -2.4922 N·m + 0.08671 N·m
= -2.4055 N·m
Torque during deceleration
[36] n_pk = omega_pk
= 942.48 rad/s
= 9000 rpm
Peak motor speed
[37] P_pk = T_pk * omega_pk
= 2.5789 N·m × 942.48 rad/s
= 2.4306 kW
Peak mechanical power
[38] t_acc = f_acc * t_move
= 0.33333 × 0.5 s
= 0.16667 s
Acceleration time
[39] t_const = t_move - 2 * t_acc
= 0.5 s − 2 × 0.16667 s
= 0.16667 s
Constant-velocity time
[40] t_cycle = t_move + t_dwell
= 0.5 s + 0.5 s
= 1 s
Cycle time

```
[41] P_rms = omega_pk * sqrt((t_acc * (2 / 3) * (T_load^2 + T_acc^2) + T_load^2 * t_const) /
= 942.48 rad/s × sqrt((0.16667 s × 2 / 3 × ((0.08671 N·m)^2 + (2.4922 N·m)^2) + (0.08671 N·m)
0.16667 s) / 1 s)
= 0.78414 kW
```
RMS mechanical power over the cycle (thermal-equivalent continuous power)
```
[42] T_rms = sqrt((T_pk^2 * t_acc + T_load^2 * t_const + T_dec^2 * t_acc) / t_cycle)
= sqrt(((2.5789 N·m)^2 × 0.16667 s + (0.08671 N·m)^2 × 0.16667 s + (-2.4055 N·m)^2 × 0.16667
= 1.4402 N·m
```
RMS torque over the cycle (thermal)
```
[43] J_ratio = (J_refl + J_mech) / J_motor
= (0.00014072 kg·m² + 0.0001 kg·m²) / 0.0002 kg·m²
= 1.2036
```
Inertia ratio (load : motor)
```
[44] u_peak = SF * T_pk / T_peak
= 1.25 × 2.5789 N·m / 6 N·m
= 0.53728
```
Peak torque utilisation with safety margin
```
[45] u_rms = SF * T_rms / T_rated
= 1.25 × 1.4402 N·m / 2 N·m
= 0.90012
```
Continuous torque utilisation with safety margin
```
[46] u_speed = n_pk / n_max
= 9000 rpm / 10000 rpm
= 0.9
```
Speed utilisation
```
[47] duty = T_rms / T_rated
= 1.4402 N·m / 2 N·m
= 0.7201
```
Thermal duty (without margin)

5. Results

SYMBOL	VALUE	DESCRIPTION
k	1.5915 mm	Transmission constant
T_load	0.08671 N·m	Load torque at motor
J_refl	0.00014072 kg·m²	Reflected load inertia
J_total	0.00044072 kg·m²	Total inertia
v_pk	1.5 m/s	Peak velocity
a_pk	9 m/s²	Peak acceleration
alpha_pk	5654.9 rad/s²	Motor angular acceleration
T_acc	2.4922 N·m	Acceleration torque
T_pk	2.5789 N·m	Peak torque
T_dec	-2.4055 N·m	Deceleration torque
T_rms	1.4402 N·m	RMS torque
n_pk	9000 rpm	Peak speed
P_pk	2.4306 kW	Peak power
P_rms	0.78414 kW	RMS power (continuous equivalent)

SYMBOL	VALUE	DESCRIPTION
J_ratio	1.2036	Inertia ratio
duty	0.7201	Thermal duty
u_peak	0.53728	Peak utilisation
u_rms	0.90012	Continuous utilisation
u_speed	0.9	Speed utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	RMS torque SF·T_rms ≤ T_rated	u_rms = 0.90012	u_rms ≤ 1	0.9	
PASS	Speed n_pk ≤ n_max	u_speed = 0.9	u_speed ≤ 1	0.9	
PASS	Peak torque SF·T_pk ≤ T_peak	u_peak = 0.53728	u_peak ≤ 1	0.537	

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: J_motor, T_rated, T_peak, n_max (Candidate motor rating). Enter each as one so its source, edition and licence travel with it.
WARNING — input.T_L: T_L (Constant load torque at the output (rotary)) was not entered; the template default 0 N*m is used.
WARNING — input.duty_mode: duty_mode (Duty (point-to-point move cycle, or continuous rotary running with a run-up)) was not entered; the template default 'point-to-point' is used.
WARNING — overhauling: T_dec: the motor brakes during deceleration (negative torque); the drive must absorb the regenerated energy.

8. Verification

Template elec.motor-sizing@1.4.0 Content hash 0f221113aba05f54 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0